

Find the general solution of each differential equation.

$$1) \frac{dy}{dx} =$$

$$2) \frac{dy}{dx} = \frac{1}{\sec^2 x}$$

$$\frac{dy}{dx} = 3e^x$$

$$3) \frac{dy}{dx} = \frac{1}{x^2}$$

$$4) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) =$$

$$5) \frac{dy}{dx} = 2e^{x-y}, y(-3) = 1$$

$$6) \frac{dy}{dx} = \frac{1}{\sec^2 x}, y(3) =$$

$$7) \frac{dy}{dx} = \frac{e^x}{x^2}, y(-1) = \frac{1}{2}$$

$$8) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) =$$

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$$11) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) =$$

$$\begin{aligned}
 &C_1 y \\
 &+ \frac{1}{3} y = \ln(2e^x + 1) \\
 &\frac{1}{3} = \frac{1}{2} \\
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 & \frac{1}{3} y + C_1 y = \ln(2e^x + 1) \\
 & y \left(\frac{1}{3} + C_1 \right) = \ln(2e^x + 1) \\
 & y = \frac{\ln(2e^x + 1)}{\frac{1}{3} + C_1}
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$$) \frac{dy}{dx} = \frac{1}{\sec}$$

$$\frac{dy}{dx} = 3e^x$$

$$) \frac{dy}{dx} = \frac{2}{\rho}$$

$$= \frac{2x}{v^2}, y(2) =$$

$$2e^{x-y}, y(-3) = 1$$

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$$\frac{e^x}{v^2}, y(-1) = \frac{1}{v}$$

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4

$$\arctan y = \frac{\pi}{4}$$

$$\tan^{-1}(y) = \frac{\pi}{4}$$

$$y = \ln(3e^x + C)$$

$$\sqrt{5-4}$$

$$e^y = 2e^x + 3$$

$$y = \tan^{-1}(x) = \frac{\ln(2x^2 + 3)}{2}$$

$$+ \frac{1}{3}y = \ln(2e^x + 3)$$

$$3y = \cos^{-1}(x) \Rightarrow \frac{e^{-y} - e^y}{2} = x^2 - \frac{1}{2}$$

$$\frac{3}{2} = \frac{1}{2}$$

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